

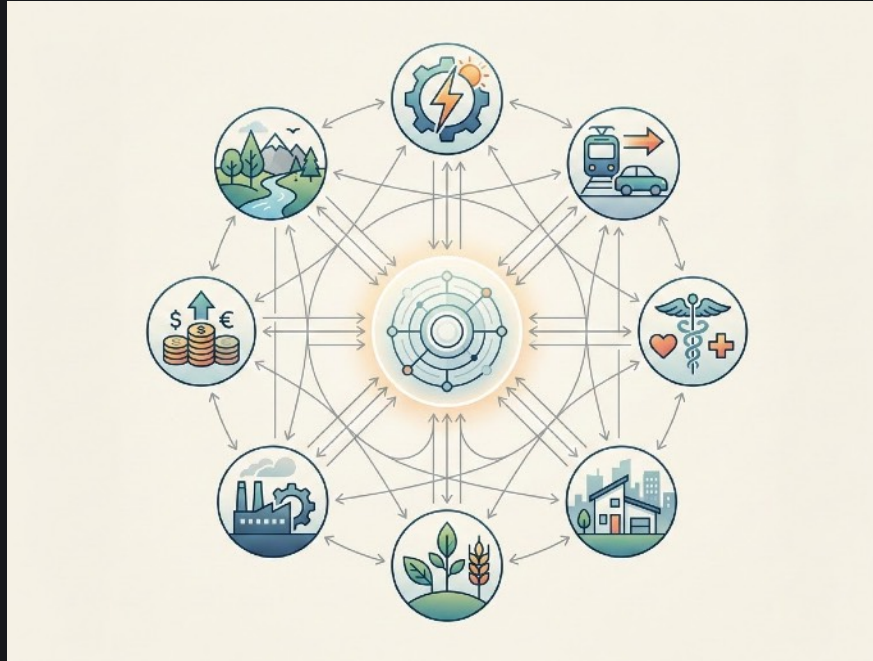


# Rethinking Development

An Integrated Trans-sectoral Perspective

*How can prosperity endure?*

# Why Trans-sectoral?



Energy • Transport • Health • Housing • Agriculture • Industry • Finance • Environment



# The Old Development Promise



*"More industrial production  
would ultimately create a  
better life for all."*

# From Scarcity to Scale

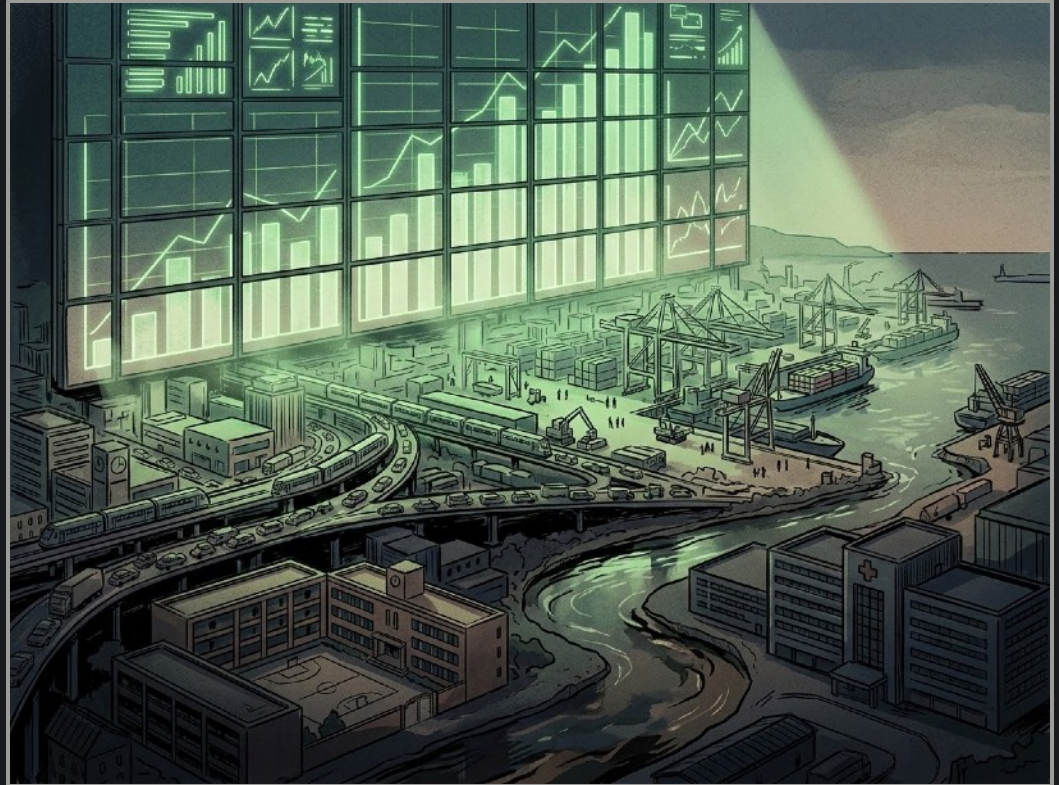
Scale lowered cost. Fossil energy accelerated scale.





# GDP: The Great Scoreboard

- Measures market production
- Tracks economic activity
- Does not equal well-being



# Growth Is Not Development

More output

Better lives

Not always the same



# What GDP Leaves Out

- Unpaid work
- Distribution and dignity
- Depletion and damage





# The Post-industrial Shift



Manufacturing

c. 1910

Services & Finance

c. 1970

Digital & Data

c. 2010

Global Logistics

c. 2024

The economy became **less visible**—not **less material**.

# The Hidden Geography of Consumption



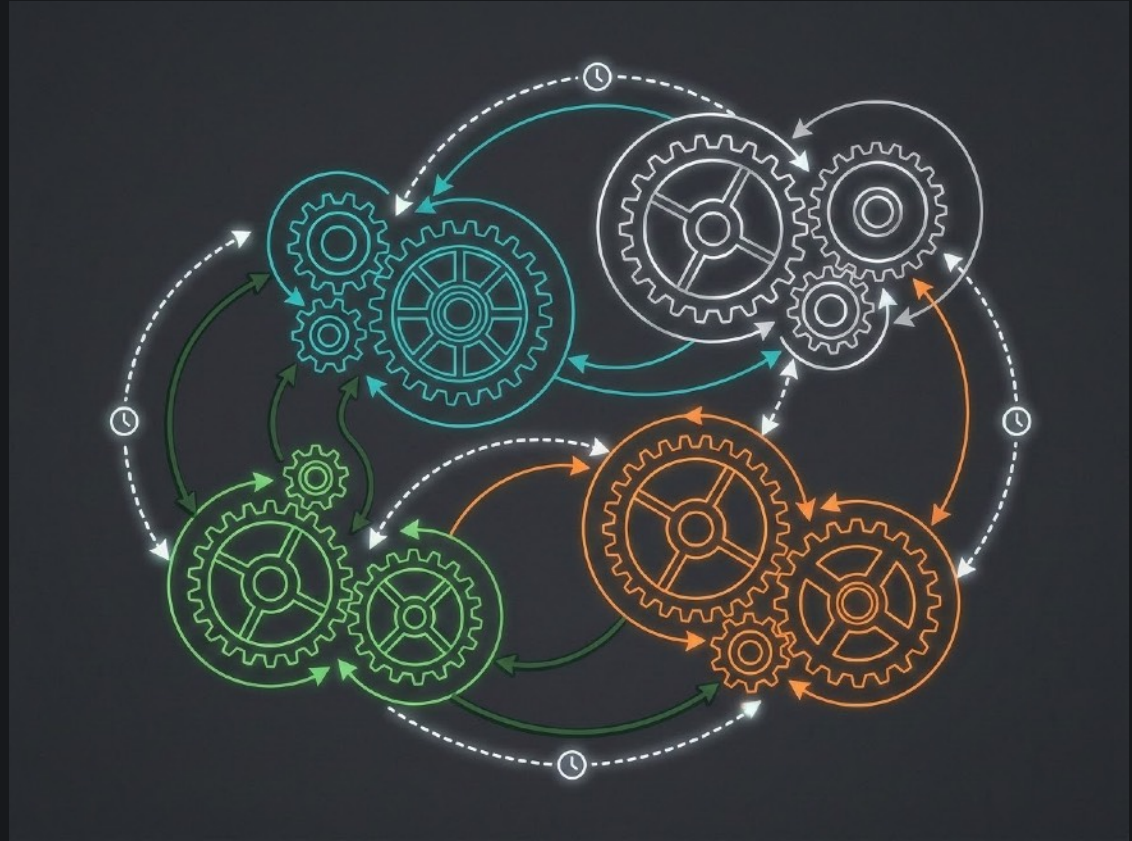
01 Extracted here

02 Made there

03 Consumed elsewhere

# Development as a Systems Problem

- Connections
- Feedback loops
- Unintended consequences





# The Economy Lives Inside Nature



The economy is a subsystem—not the whole system.

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# Nature as Source

- Materials
- Energy
- Living resources



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# Nature as Sink

- Air receives emissions
- Water receives effluent
- Land receives waste





# Nature as Life-support

Service 01 Pollination

Service 02 Flood control

Service 03 Soil fertility

Service 04 Climate  
regulation

# Ecosystem Services



## Provisioning

Material or energy outputs from ecosystems:

- Food & crops
- Freshwater
- Timber & wood
- Medicinal plants



## Regulating

Services that regulate ecological processes:

- Local climate
- Carbon storage
- Flood control
- Water purification



## Cultural

Non-material benefits obtained from nature:

- Recreation
- Aesthetic value
- Spiritual heritage
- Education



## Supporting

Underpinning processes for all other services:

- Soil formation
- Nutrient cycling
- Water cycling
- Photosynthesis

# Why Environmental Goods Are Different

**No clear owner.**

**No easy boundary.**

**No complete price.**

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# Four Types of Goods

RIVAL

NON-RIVAL

EXCLUDABLE

## Private Goods

Food, clothing, car

## Club Goods

Toll roads, cinemas, cable TV

NON-EXCLUDABLE

## Common Resources

Fish stocks, timber, groundwater

## Public Goods

Clean air, national defense



# Public Goods & Free-riding

- Everyone benefits
- Few volunteer to pay

Takeaway

**Result: under-provision**



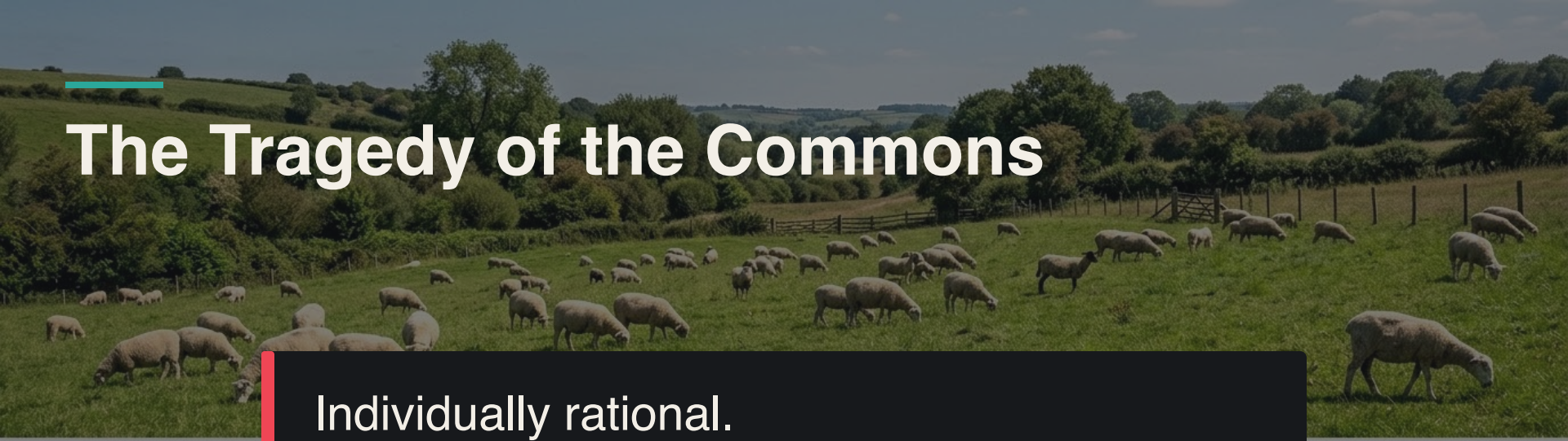
# Common Resources

- Private gain from extraction
- Shared cost of depletion
- Pressure to overuse



**Case Study** Exploitation of marine fisheries as a classic common-pool resource.





# The Tragedy of the Commons

Individually rational.  
**Collectively disastrous.**

# When Markets Fail

## SPILLOVERS & EXTERNAL EFFECTS

Missing Costs



Missing Information

Missing Institutions



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# When Prices Lie

- Pollution not charged
- Health damage displaced
- Resources treated as free



# Externality

Private decision

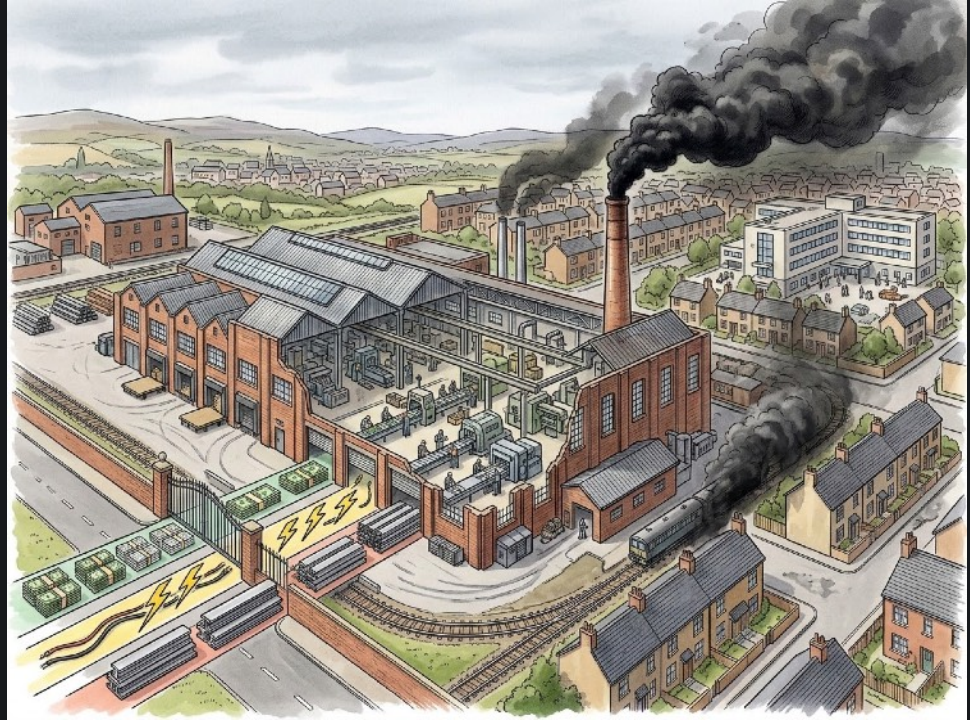
Third-party effect

Missing from price



# Negative Production Externality

- Firm counts private cost
- Society bears added harm
- Output becomes excessive



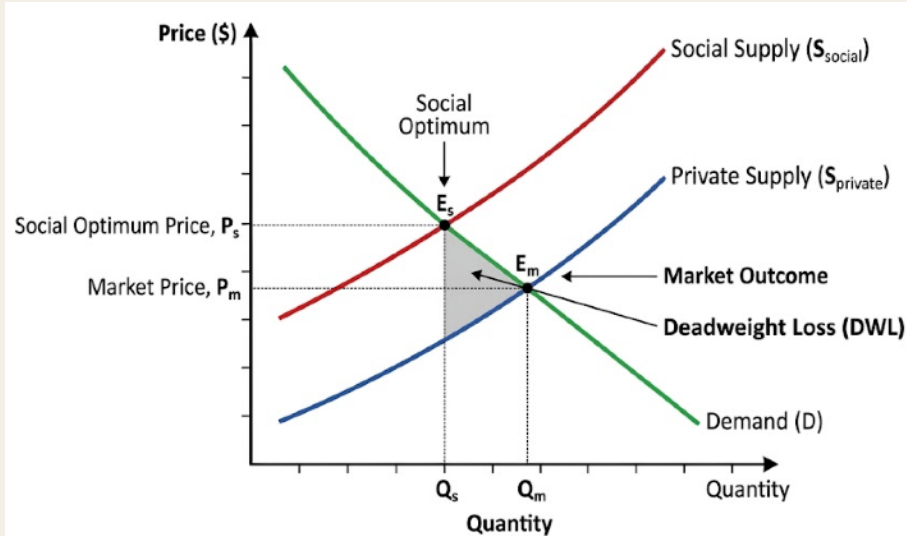


# Private Cost vs Social Cost



$$\text{Social Cost} = \text{Private Cost} + \text{External Cost}$$

# Pollution Creates Deadweight Loss



**System Inefficiency**

**Market output exceeds the social optimum.**

# Positive Externalities

## Economic Dynamics

- Private action
- Wider social benefit
- Market supplies too little





# Correcting Market Failure



**Tax**



**Subsidy**



**Standard**



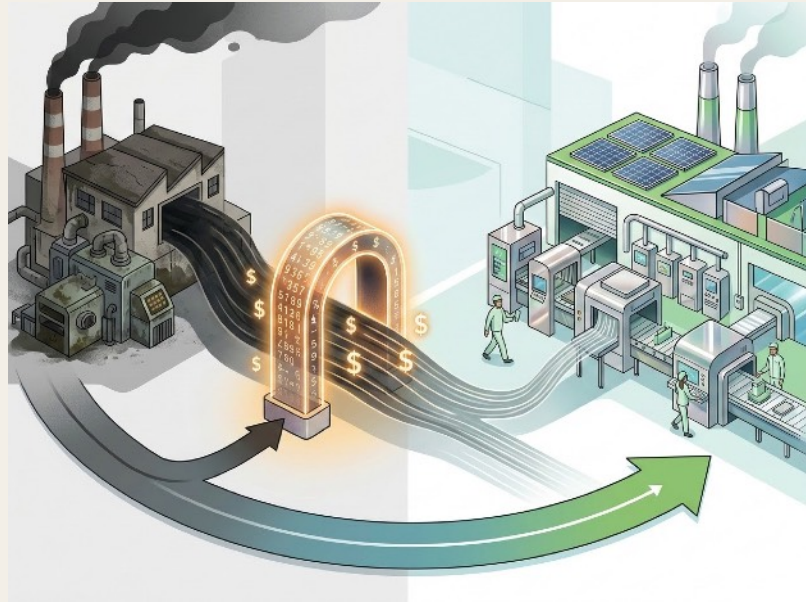
**Permit**

# Pigouvian Tax

- Price the external damage
- Change private incentives

Core Principle

**Polluter Pays**



# Subsidies for Clean Choices

- Lower transition cost
- Reward social benefits
- Avoid permanent dependence



# The Coase Approach

Prerequisites for Negotiation

**01. Clear rights**

**02. Low transaction costs**

**03. Few affected parties**





# Which Policy Instrument?



Standards

Certainty



Taxes

Flexibility



Permits

Quantity  
Control



Liability

Accountability

# Bhopal: Risk Was Underpriced

01

Private savings

02

Public risk

03

Long-term harm

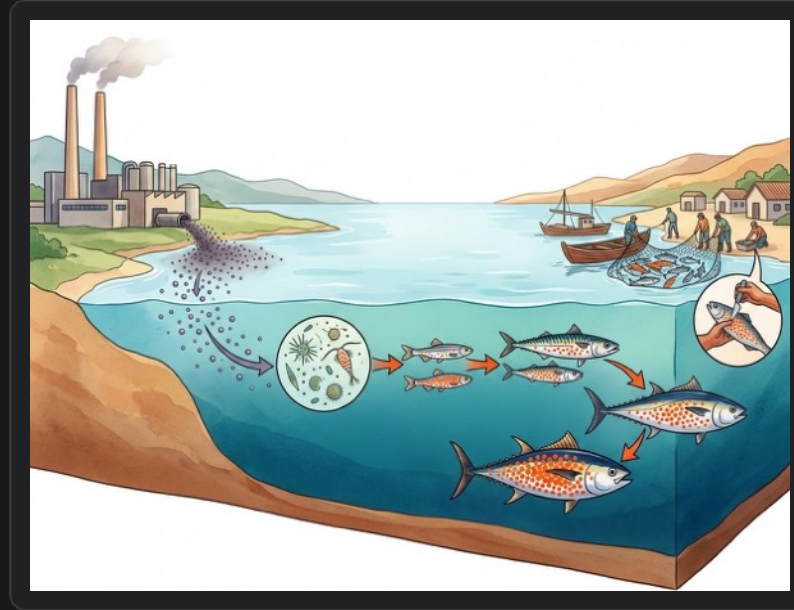


# The Economics of Bhopal



**Prevention looked costly—until failure occurred.**

# Minamata: Pollution Through Food

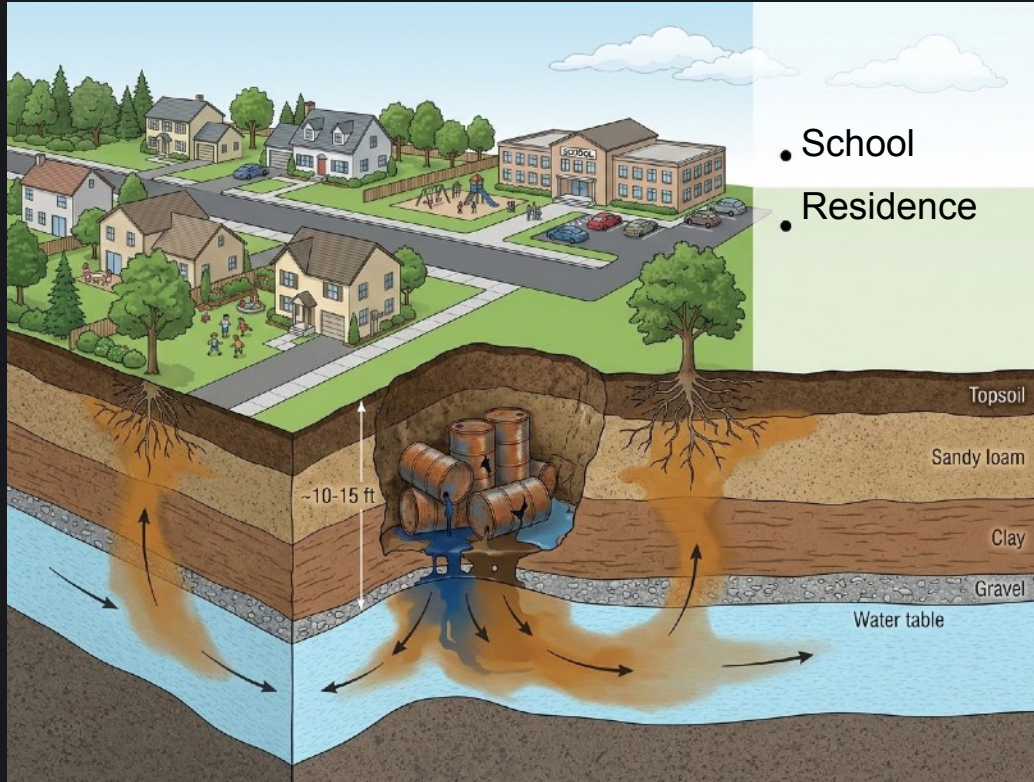


- **Mercury discharge**

- **Bioaccumulation**

- **Delayed disease**





# Love Canal: Toxic Legacy

## Phase 01

Waste buried and forgotten

## Phase 02

Housing followed

## Phase 03

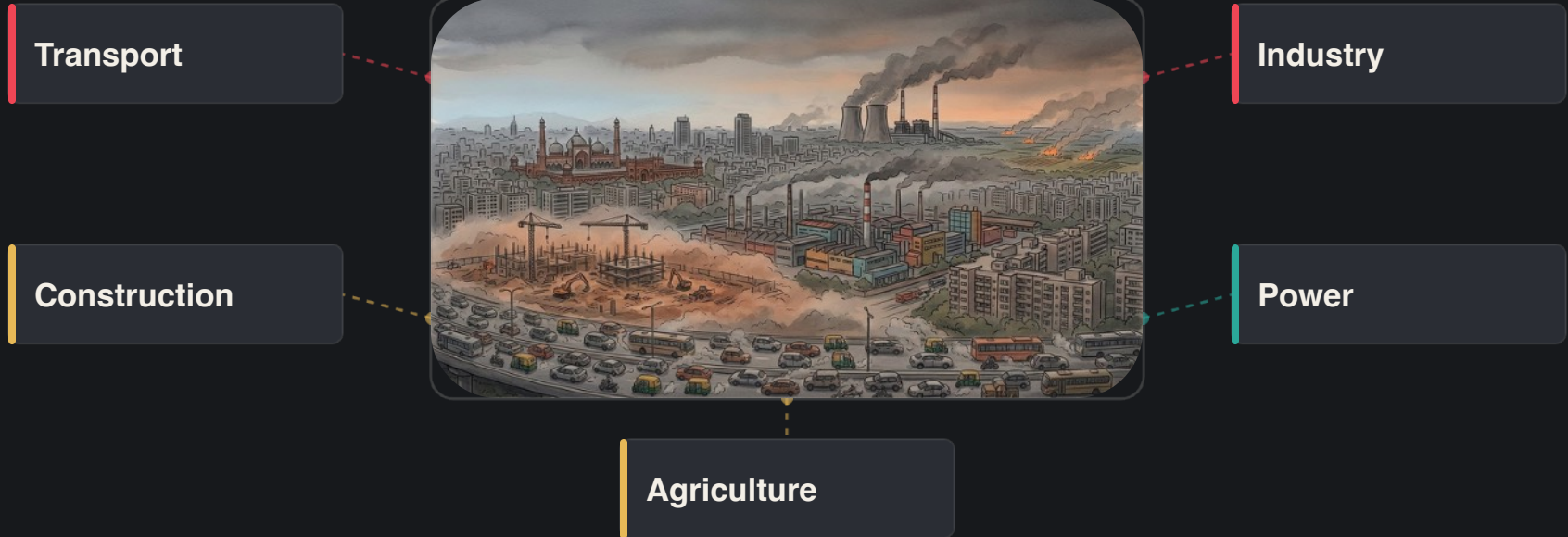
Risk returned decades later

# Environmental Justice



Who receives the **benefits**—and who bears the **costs**?

# Delhi Air Is Not One Problem





# Urbanization - Engine of Development

## Economic Advantages

- Productivity through proximity
- Specialization and innovation
- Shared infrastructure





# Urbanization: Concentration of Risk

Risk 01  
**Congestion**

Risk 02  
**Heat**

Risk 03  
**Flooding**

Risk 04  
**Waste**

# The Urban Metabolism





# Urban Heat Island

- Concrete stores heat
- Trees and water cool
- Heat burden is unequal





# Designing Sustainable Cities

## Planning Pillars

- Compact and connected
- Green-blue infrastructure
- Integrated services





# Green Buildings

- 01 **Passive design**
- 02 **Efficient systems**
- 03 **Water reuse**
- 04 **Low-impact materials**



# The Sustainability Question

## Present Infrastructure

Heavy resource consumption, built environments, and active industrial systems footprint.



## Future Landscape

Regenerative design, pristine ecosystems, and preserved choices for tomorrow.

**Can today's development preserve tomorrow's choices?**

# How Sustainability Emerged



01

Environmental  
awakening



02

Stockholm



03

Brundtland



04

Rio



05

SDGs



# The Brundtland Idea



- **Present needs**
- **Future capability**
- **Priority to the poor**

# Two Dimensions of Equity



## Within this generation

(Who bears the burden?)



## Across generations

(Who inherits the cost?)

# Triple Bottom Line





# The SDGs as a Network



**Progress on one goal reshapes the others.**

# Substitutable Capital

## Man-Made Alternatives

Infrastructure, machinery, and financial assets that can be engineered, traded, and fully replaced.



# Critical Natural Capital

## Irreplaceable Biosphere

Essential ecosystem services that lack engineering substitutes. Human-made assets cannot replace critical life-support networks.



# Dynamic Limits

Dynamic limits define ecological operating thresholds.

- Resource regeneration
- Waste absorption
- Ecosystem resilience





# Damage may accelerate after a threshold.



I. PRISTINE FLOW



II. ABSORPTION LIMIT



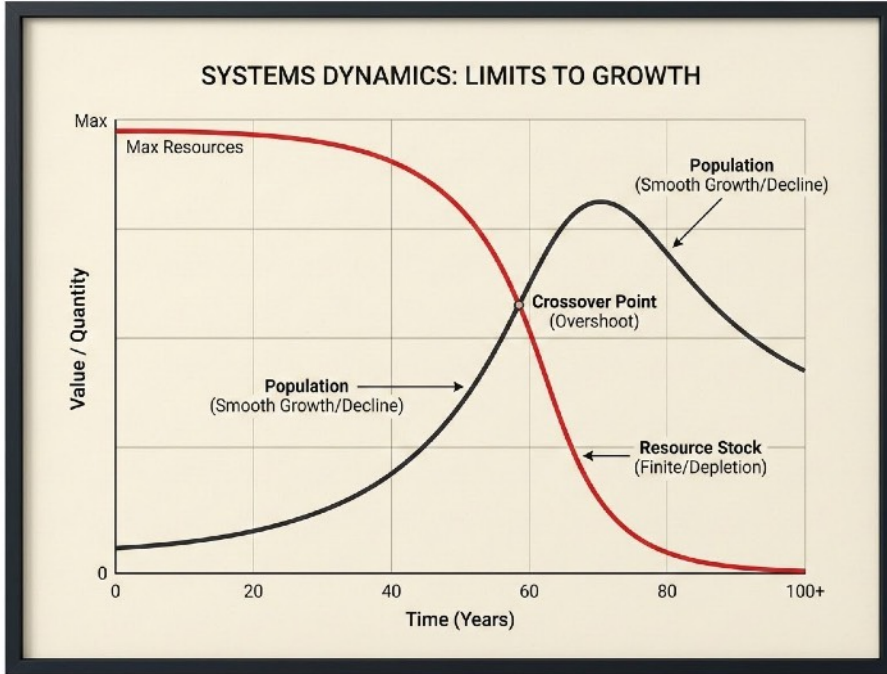
III. SYSTEM OVERLOAD



Non-linear ecological responses occur when waste exceeds natural absorption capacity.



# Limits to Growth



Dynamics 01

**Exponential growth**

Dynamics 02

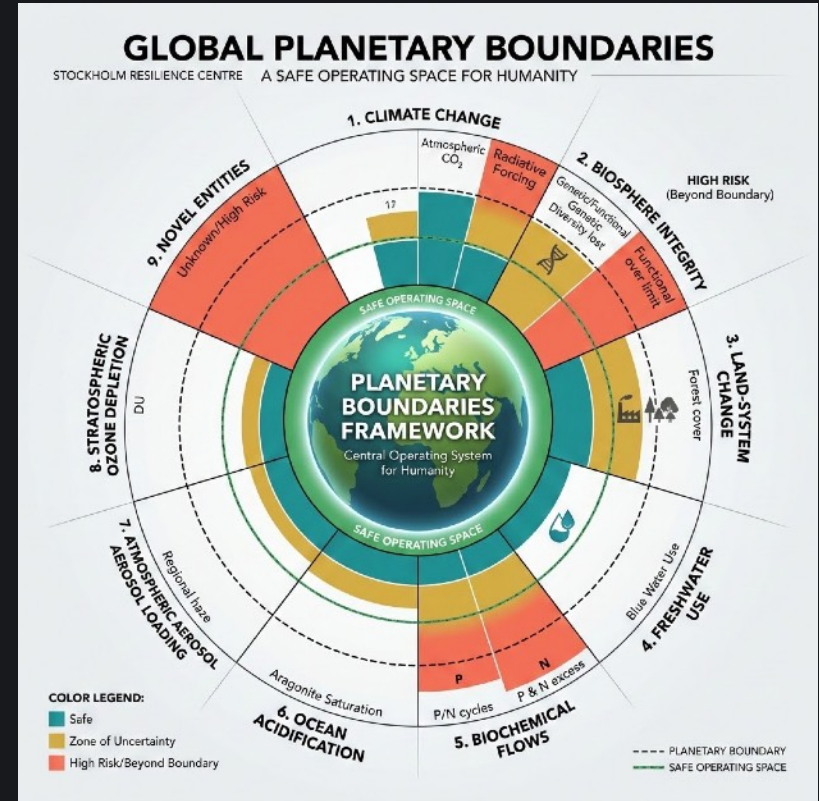
**Finite stocks.  
Delayed feedback.**

Dynamics 03

**Unchecked expansion in  
bounded systems can lead to  
overshoot.**

# A safe operating space, not a precise cliff edge.

Interacting Earth-system  
constraints guide sustainable  
human development.



# Tracking Ecological Demand

## Tracking aggregate ecological demand

- Demand on biocapacity
- Regeneration has limits
- Consumption drives scale



# The I=PXAXT Equation

Demographic scale, consumption levels, and technological intensity drive environmental change.

$$\text{Impact} = \text{Population} \times \text{Affluence} \times \text{Technology}$$

P POPULATION



Demographic Scale

A AFFLUENCE



Consumption Levels

T TECHNOLOGY



Technological Intensity



# Scale increases baseline environmental pressure.

- More people, more total demand
- Density can also create efficiency



Scale of Population Demand

Transit-Efficient Development

# Affluence: The Consumption Effect

Rising income transitions household demand from survival to high-intensity luxury.



## STEP 01

Basic Needs

## STEP 02

Comfort

## STEP 03

High Material Demand

# TECHNOLOGY & SYSTEMIC SCALE

## FOSSIL FUELS

Traditional extraction achieves high density but imposes massive ecological costs.



## RENEWABLES

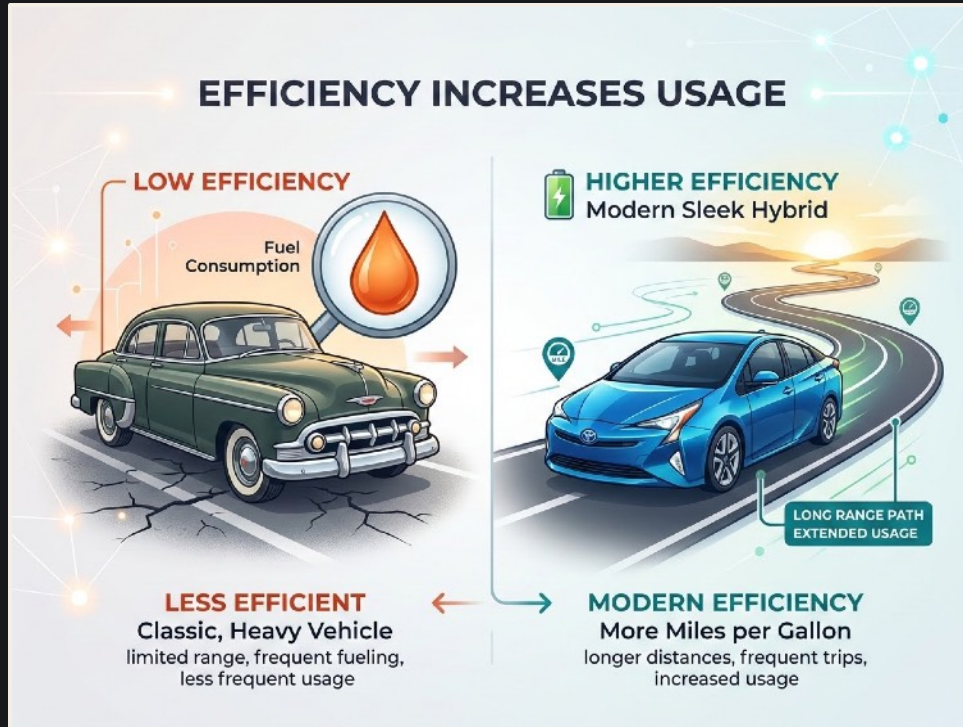
Clean infrastructure lowers intensity but demands greater land and materials.

**Does technology reduce impact — or expand throughput?**

Innovation lowers intensity but often enables greater absolute scale.



# Technical savings are often offset by behavioral changes.



## Phase 01

Efficiency rises.

## Phase 02

Operating cost falls.

## Phase 03

Use may increase.



# The Production Cost Ledger



Hidden externalities must be accounted for.

Internalized Business Costs

Labour + Materials + Energy

Externalized Social Costs

**+ Health + Ecology + Risk**

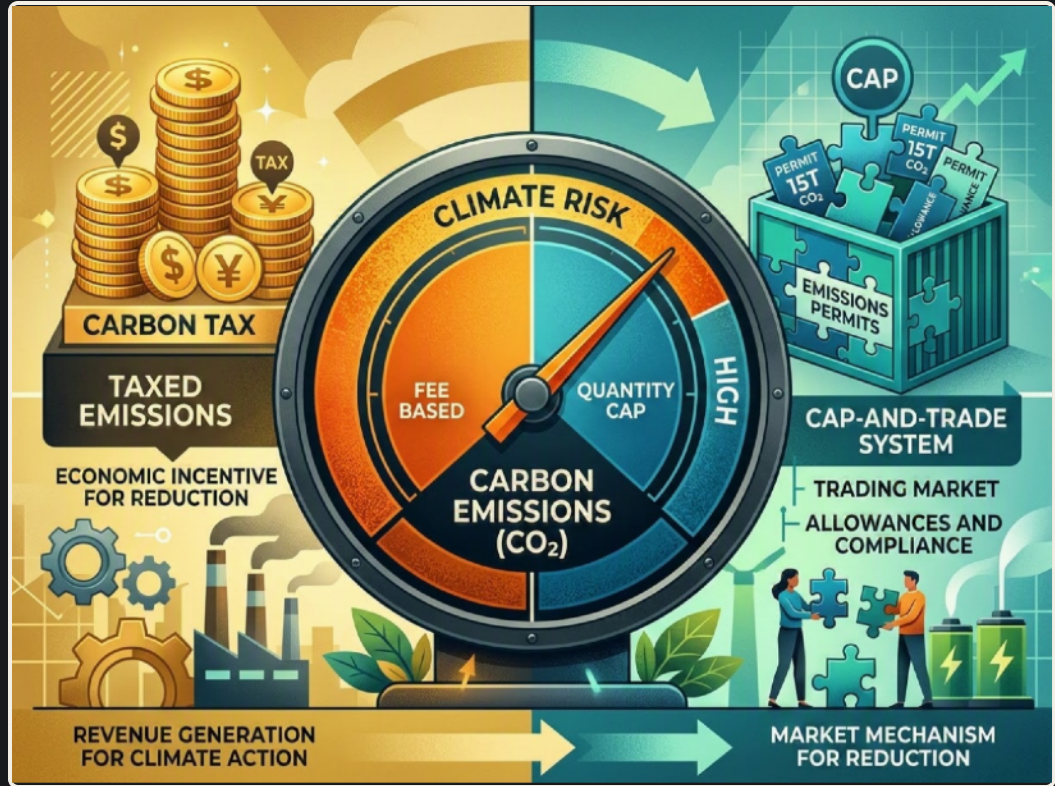
**TRUE COST = (Labour + Materials + Energy) + (Health + Ecology + Risk)**

Dr. Ankur Awadhiya, IFS

# Market Mechanisms

Market mechanisms internalize the true cost of emissions.

- **Tax** sets the price
- **Cap** sets the quantity



# The Linear Economy Pipeline

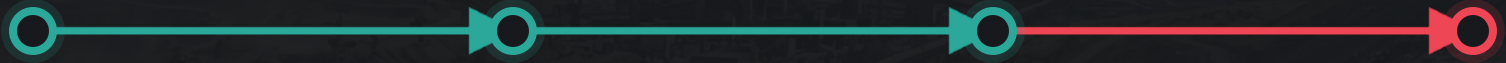


01 / TAKE  
EXTRACTION

02 / MAKE  
PRODUCTION

03 / USE  
CONSUMPTION

04 / DISPOSE  
LANDFILL



**Value moves forward. Waste accumulates behind.**

The take-make-dispose model generates unsustainable resource depletion and pollution.

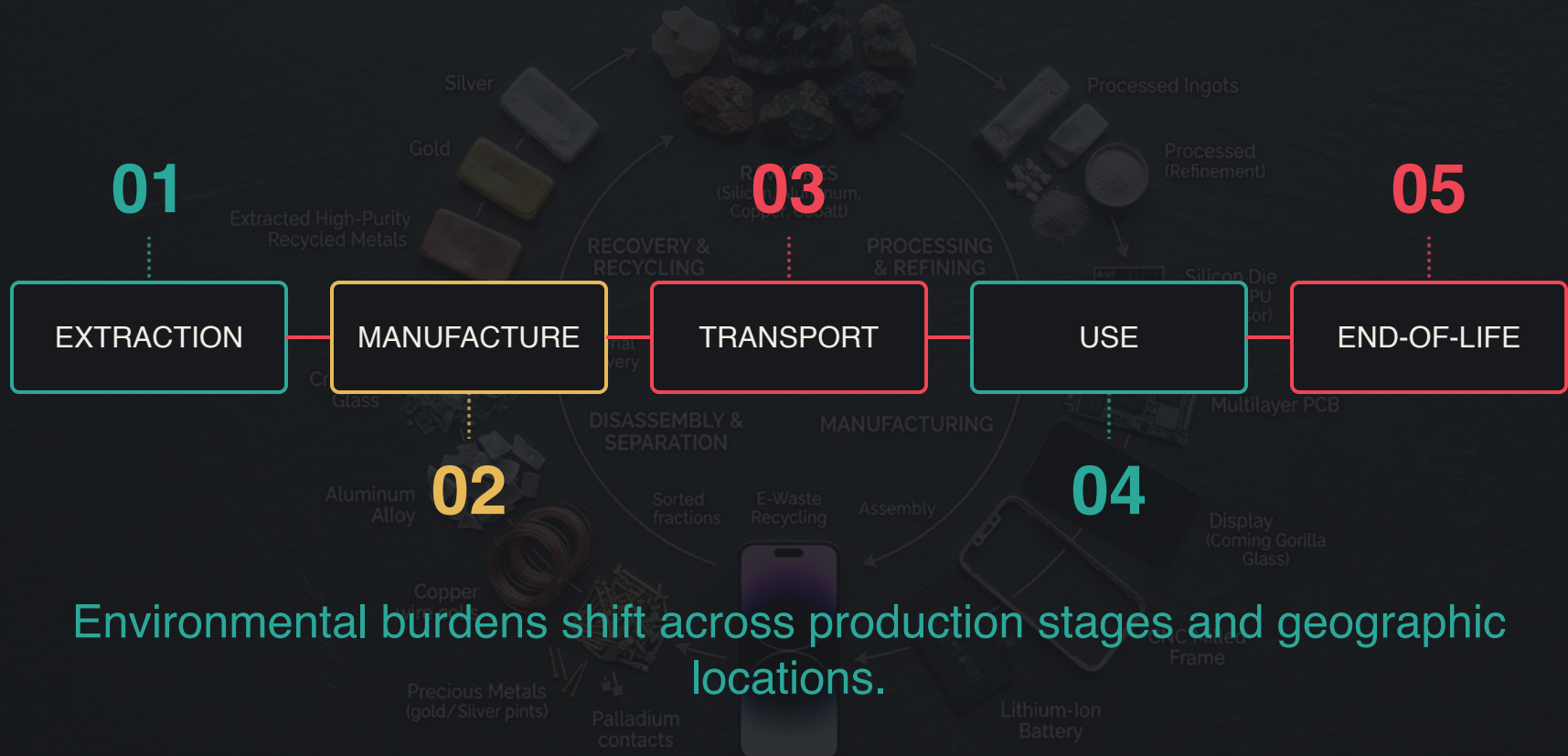


# A Hierarchy of Continuous Value Retention



Design out waste • Keep products in use • Regenerate natural systems

# Smartphone Life-Cycle Analysis



# Responsibility After Sale

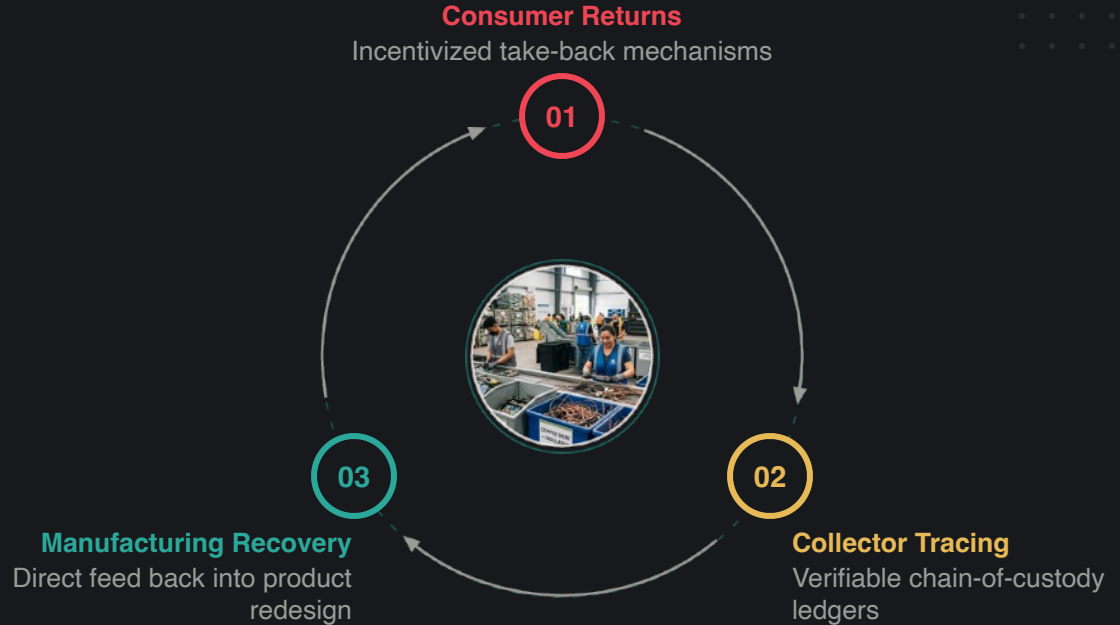
Incentivizing post-consumer accountability and redesign.

## ■ Take-Back & Traceability

Establishing strict custody tracking from consumers back to recovery centers.

## ■ Design for Recovery

Redesigning architectures to facilitate seamless material sorting and reuse.



# Balancing Human Needs with Planetary Boundaries

## Ecological Ceiling

Planetary boundaries that define the critical limits of our global ecosystem. Exceeding these thresholds increases systemic risk.

## Safe and Just Space

The balanced, regenerative corridor where humanity can thrive sustainably, reconciling basic rights with physical boundaries.

## Social Foundation

The baseline of fundamental human needs—such as food security, health, equity, and housing—that must be secured for all people.





Not less  
development.

**Better development.**

Redesigning progress for  
thriving ecosystems and  
equitable human well-being.

